A retraction between functions induces retractions of all their fibers

The Univalent Foundations Program

hott-book:equivalences.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author's 'lem:func_retract_to_fiber_retract' (source lines 677-680, Lemma 4.7.3); the proof environment starts on line 682, after \end{lem} and one blank line, so the pairing is by adjacency.

Editorial note. The retained context is the author's definition of a function being a retract of another, which both the statement and the proof invoke by label; retaining it keeps that label defined inside the package.

Editorial note. The retained counter values (chapter 4, section 7, theorem offset 1) make the retained definition print as Definition 4.7.2 and the counted lemma as Lemma 4.7.3, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. This source directory intentionally contains only the author file, so the retained book macro library is bundled as a per-task copy inside this task directory and referenced from preamble_spans; the copy is verbatim from the same pinned book snapshot and no macro is added, changed or invented.

Editorial scope. One original result of the HoTT Book, counted once: retractions of functions induce retractions of their fibers (Lemma 4.7.3).

Allowed standard prerequisites (editorial).

- Retract of a function. g : A -> B is a retract of f : X -> Y when there are s : A -> X, r : X -> A, L : prd a, f(s a) = s'(a), K : prd x, r'(f x) = g(r x), and the coherence data recorded in the author's definition.. All these maps and homotopies are used to write phi_b and psi_b at source lines 685-688.
- Fiber of a map. hfiber g b is the type of pairs (a, g a = b).. It is the type on both sides of the retraction.
- Omission of contractible components. A sigma type whose first component ranges over a contractible type is equivalent to the fiber of the remaining component.. It is the reformulation applied at source line 696.

Source reference. Original theorem 3.11.9: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1844–1844. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is

replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Source reference. Original theorem 2.7.2: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1853–1853. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Definition 4.7.2. A function $g : A \to B$ is said to be a **retract** of a function $f : X \to Y$ if there is a diagram

$$\begin{array}{ccccc} A & \xrightarrow{s} & X & \xrightarrow{r} & A \\ {\scriptstyle g}\downarrow & & {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle g} \\ B & \xrightarrow[s']{} & Y & \xrightarrow[r']{} & B \end{array}$$

for which there are

(i) a homotopy $R : r \circ s \sim \mathsf{id}_A$.
(ii) a homotopy $R' : r' \circ s' \sim \mathsf{id}_B$.
(iii) a homotopy $L : f \circ s \sim s' \circ g$.
(iv) a homotopy $K : g \circ r \sim r' \circ f$.
(v) for every $a : A$, a path $H(a)$ witnessing the commutativity of the square

$$\begin{array}{ccc} g(r(s(a))) & \overset{K(s(a))}{=\!=} & r'(f(s(a))) \\ {\scriptstyle g(R(a))}\Vert & & \Vert{\scriptstyle r'(L(a))} \\ g(a) & \underset{R'(g(a))^{-1}}{=\!=} & r'(s'(g(a))) \end{array}$$

Lemma 4.7.3. *If a function $g : A \to B$ is a retract of a function $f : X \to Y$, then $\mathsf{fib}_g(b)$ is a retract of $\mathsf{fib}_f(s'(b))$ for every $b : B$, where $s' : B \to Y$ is as in Definition 4.7.2.*

Proof. Suppose that $g : A \to B$ is a retract of $f : X \to Y$. Then for any $b : B$ we have the functions

$$\begin{aligned} &\varphi_b : \mathsf{fib}_g(b) \to \mathsf{fib}_f(s'(b)), && \varphi_b(a,p) :\equiv (s(a), L(a) \cdot s'(p)), \\ &\psi_b : \mathsf{fib}_f(s'(b)) \to \mathsf{fib}_g(b), && \psi_b(x,q) :\equiv (r(x), K(x) \cdot r'(q) \cdot R'(b)). \end{aligned}$$

Then we have $\psi_b(\varphi_b(a,p)) \equiv (r(s(a)), K(s(a)) \cdot r'(L(a) \cdot s'(p)) \cdot R'(b))$. We claim ψ_b is a retraction with section φ_b for all $b : B$, which is to say that for all $(a,p) : \mathsf{fib}_g(b)$ we have $\psi_b(\varphi_b(a,p)) = (a,p)$. In other words, we want to show

$$\prod_{(b:B)} \prod_{(a:A)} \prod_{(p:g(a)=b)} \psi_b(\varphi_b(a,p)) = (a,p).$$

By reordering the first two Πs and applying a version of original theorem 3.11.9, this is equivalent to

$$\prod_{a:A} \psi_{g(a)}(\varphi_{g(a)}(a, \mathsf{refl}_{g(a)})) = (a, \mathsf{refl}_{g(a)}).$$

For any a, by original theorem 2.7.2, this equality of pairs is equivalent to a pair of equalities. The first components are equal by $R(a) : r(s(a)) = a$, so we need only show

$$R(a)_*(K(s(a)) \cdot r'(L(a)) \cdot R'(g(a))) = \mathsf{refl}_{g(a)}.$$

But this transportation computes as $g(R(a))^{-1} \cdot K(s(a)) \cdot r'(L(a)) \cdot R'(g(a))$, so the required path is given by $H(a)$. □